EE1060: Differential Equations and Transform Techniques

Practice Problem Set - 05

Practice problems in topics covered till: Lecture 18

Note : The practice problem sets are designed for your personal review and are not required for submission or evaluation. These problems aim to reinforce your understanding of the material and help you prepare effectively. Some of the practice problems are sourced from the course references, and it is strongly recommended that you consult these references for additional practice problems. Due credit is also given to the TAs of the course, who have contributed to the creation of this problem set.

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1. Determine the general solution of the second-order ordinary differential equation $\left(1-x^2\right)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 6y = 0$, given that one of the fundamental solutions is $y_1 = \frac{1}{2}\left(3x^2-1\right)$.

2. Determine the general solution of the second-order ordinary differential equation $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} - 4y = 0$, given that one of the fundamental solutions is $y_1 = x^2$. [?]

3. Consider the Bessel's equation [?]
$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + \left(x^2 - p^2\right)y = 0 \tag{1}$$
with $p = \frac{1}{2}$.
   (a) Show that $y_1 = \dfrac{\sin x}{\sqrt{x}}$ is one solution over $\mathbb{R}^{++}$.
   (b) Compute the other fundamental solution.

4. Find the general solution of the second-order ODE
$$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = e^{-2x} \tag{2}$$

5. Find the general solution of the second-order ODE
$$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = x^2 \tag{3}$$

6. Find the general solution of the second-order ODE
$$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = e^{2x}\sin(4x) \tag{4}$$

7. Find the general solution of the second-order ODE
$$\frac{d^2y}{dx^2} + k^2y = \sin(kx) \tag{5}$$

8. Find the general solution of the second-order ODE

$$\frac{d^2y}{dx^2} + 4k\frac{dy}{dx} + k^2y = \sin(kx) \tag{6}$$

9. Determine the general solution of the second-order ordinary differential equation $\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 2y = 0$, given that one of the fundamental solutions is $y_1 = 2x$.

10. Find the general solution of the second-order ODE

$$\frac{d^2y}{dx^2} - y = xe^x \tag{7}$$

11. Determine the value of λ for which the boundary value problem

$$\frac{d^2y}{dx^2} + \lambda y = 0 \text{ with } y'(0) = 0 \text{ and } y'(L) = 0 \tag{8}$$

has nontrivial solution (Neumann eigenvalue problem).

12. Determine the eigenvalues and eigenfunctions of [**?**]

$$\frac{d^2y}{dx^2} + \lambda y = 0 \text{ with } y'(-\pi) = 0 \text{ and } y'(\pi) = 0 \tag{9}$$

13. Determine the value of λ for which the boundary value problem [**?**]

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + \lambda y = 0 \text{ with } y(0) = y(1) = 0 \tag{10}$$

has nontrivial solution (periodic eigenvalue problem).

14. Find the general solution of the ODE

$$y^{(3)} - 6y'' + 9y' = 3x^2 \tag{11}$$

Please Note: Some of the problems presented in this document have been adapted from textbooks referenced for this course. These problems are included to help students practice and understand key concepts. The material is used for educational purposes only. We have made every effort to provide proper citations, but if any are inadvertently omitted, it was not intentional.